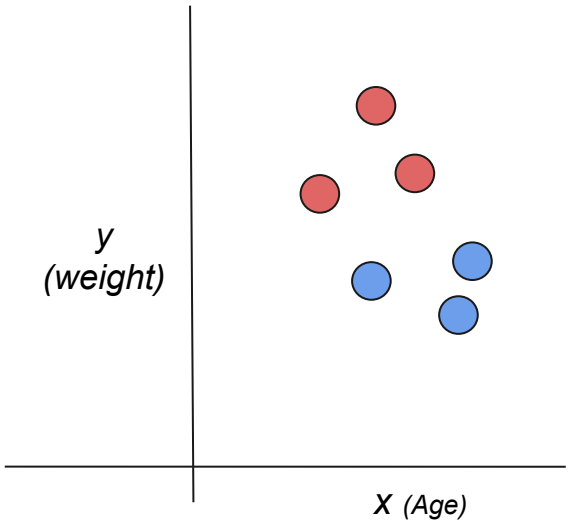
# CIS 635 Knowledge Discovery & Data Mining

## Classification Models: Logistic Regression

# Classification Models

- *k-NN*
- *Decision Tree*
- **Logistic Regression**
- Random Forest Classifier
- Support Vector Machines (SVMs)
- Boosting Classifiers
- Naive Bayes

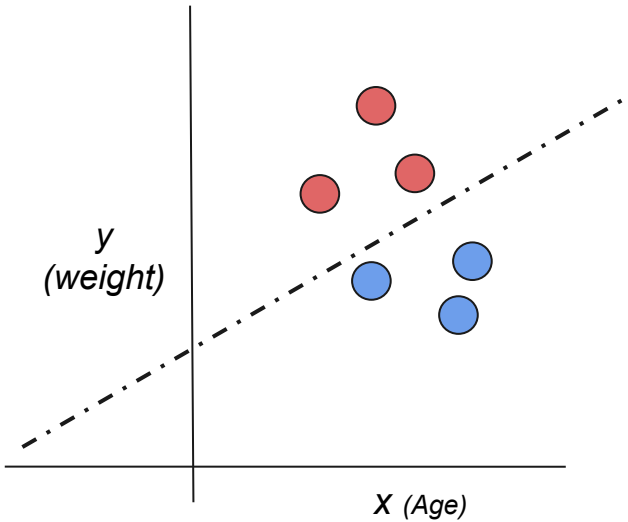
# Linear Line as a Threshold




**How to classify ??**

$y \in \{underweight(0), overweight\ (1)\}$

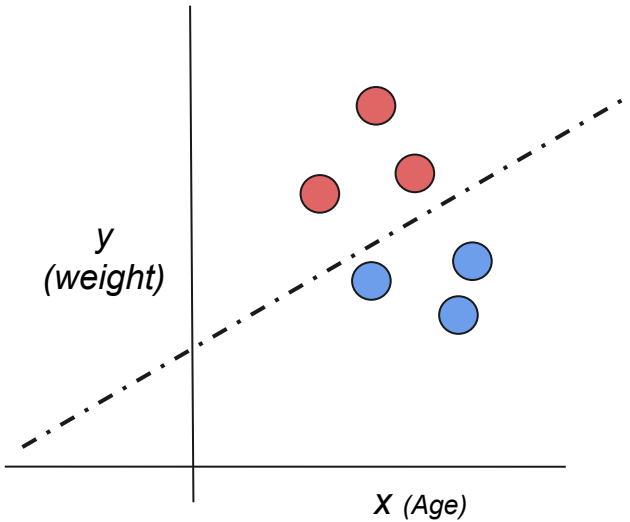
# Linear Line as a Threshold




**Straight Line as the Separator**

$y \in \{underweight(0), overweight\ (1)\}$

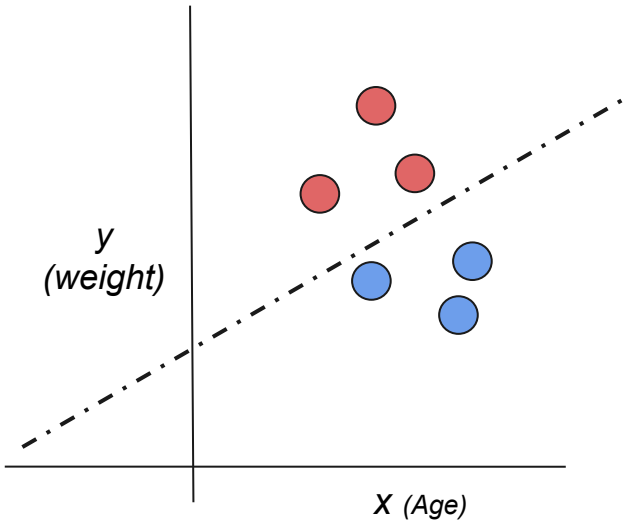
# Linear Line as a Threshold




**Straight Line as the Separator**

$$\hat{y} = \beta_0 + \beta_1 x$$

# Linear Line as a Threshold




**Straight Line as the Separator**

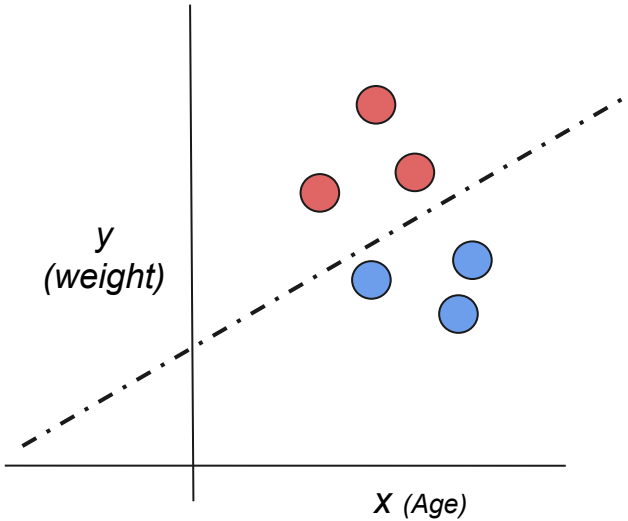
$$\hat{y} = \beta_0 + \beta_1 x$$

**Classification Rule**

$$\check{y} = \begin{cases} 1, & \text{if} \quad y - (\beta_0 + \beta_1 x) > 0 \\ 0, & \text{otherwise} \end{cases}$$

# Logistic Regression

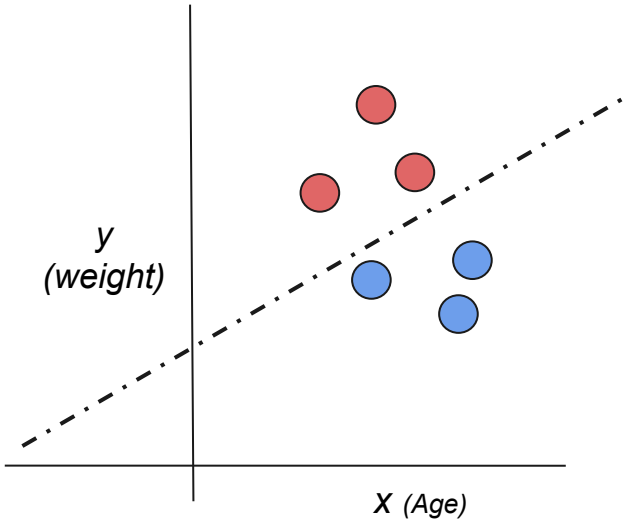
## From Regression to Classification




$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

# Logistic Regression

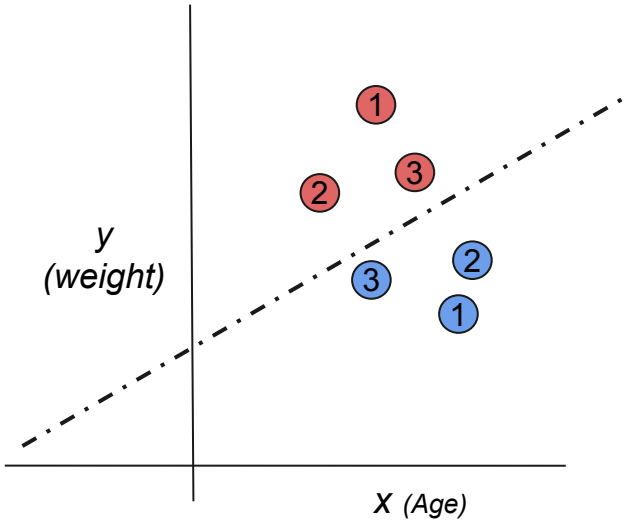
## From Regression to Classification




- Relationship to Decision Boundary
  (The dotted line in this case)

# Logistic Regression

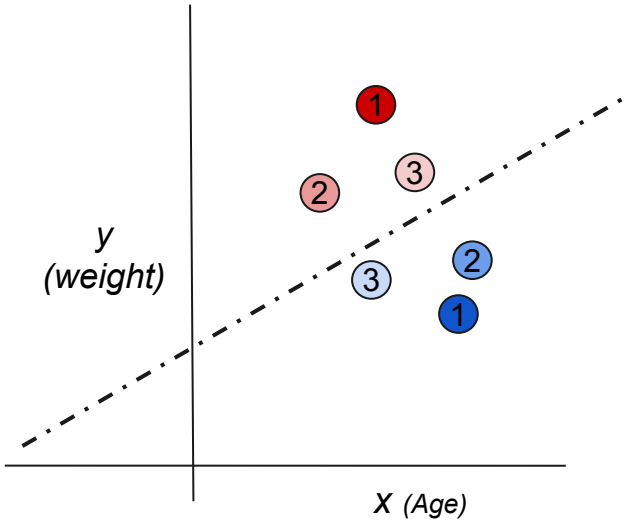
## From Regression to Classification




- Relationship to Decision Boundary (The dotted line in this case)
- We index based on their distance

# Logistic Regression

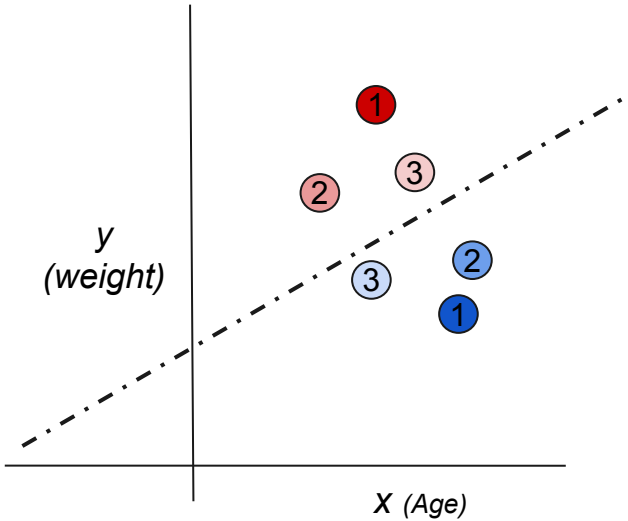
## From Regression to Classification




- Relationship to Decision Boundary (The dotted line in this case)
- We index based on their distance
- Also add color code for better understanding

# Logistic Regression

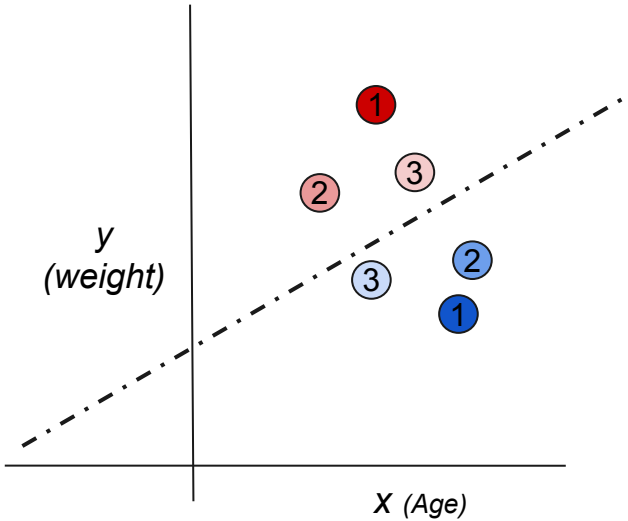
## From Regression to Classification




- Relationship to Decision Boundary (The dotted line in this case)
- We index based on their distance
- Also add color code for better understanding
- Let's do feature encoding

# Logistic Regression

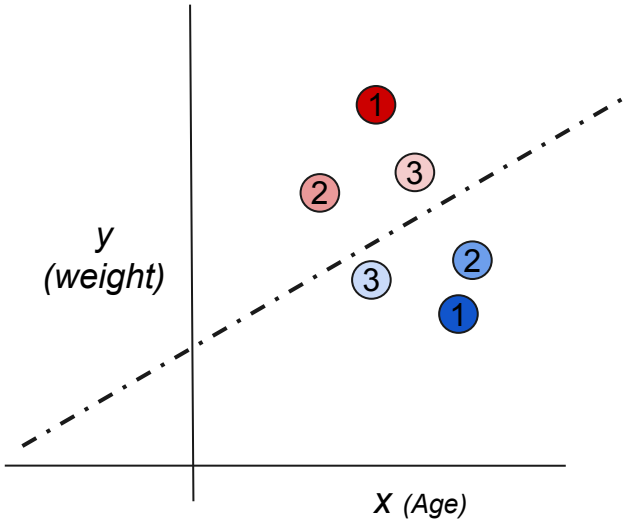
## From Regression to Classification



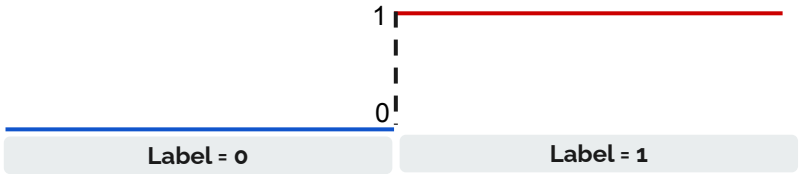

- Let's do feature encoding: ***Binary {0, 1}***

# Logistic Regression

## From Regression to Classification

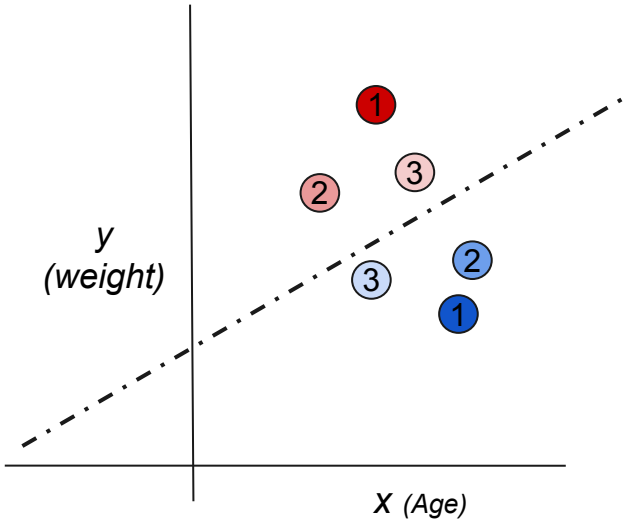



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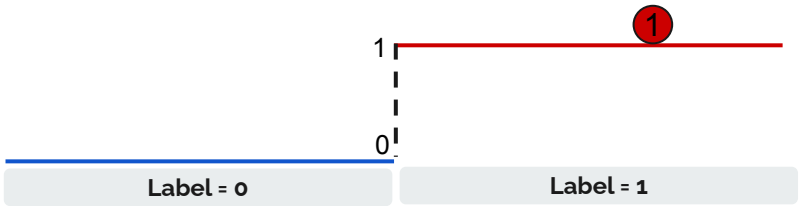


# Logistic Regression

## From Regression to Classification

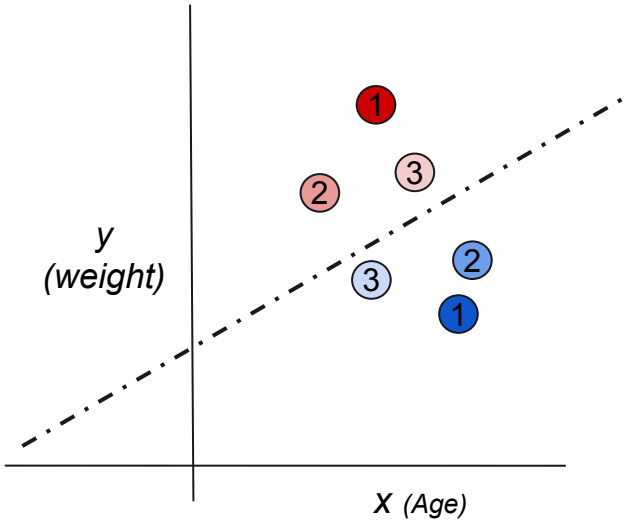



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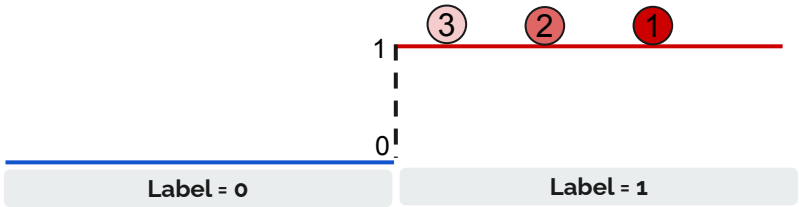


# Logistic Regression

## From Regression to Classification

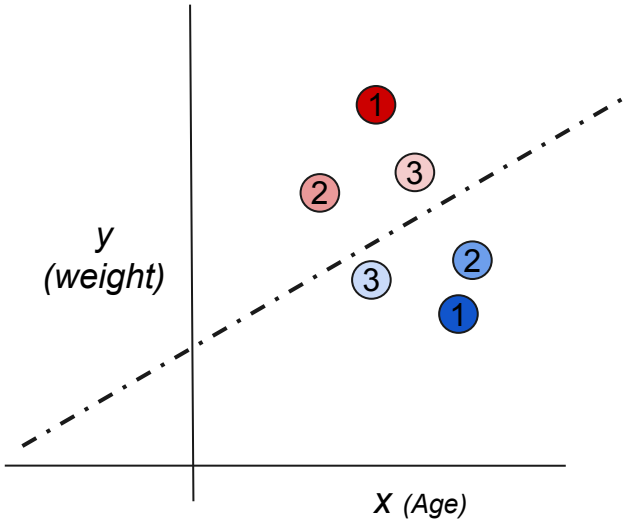



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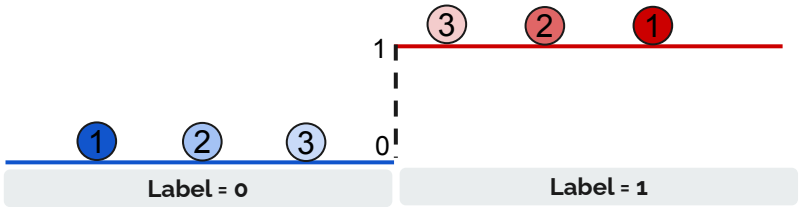


# Logistic Regression

## From Regression to Classification

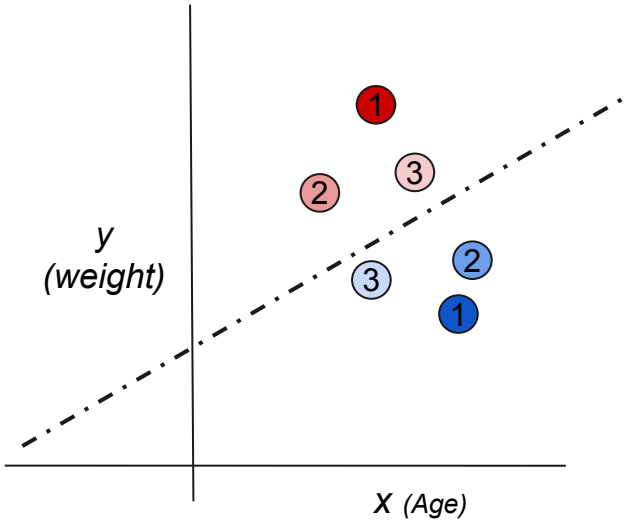



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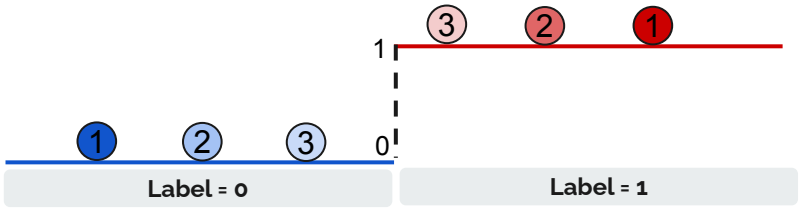


# Logistic Regression

## From Regression to Classification

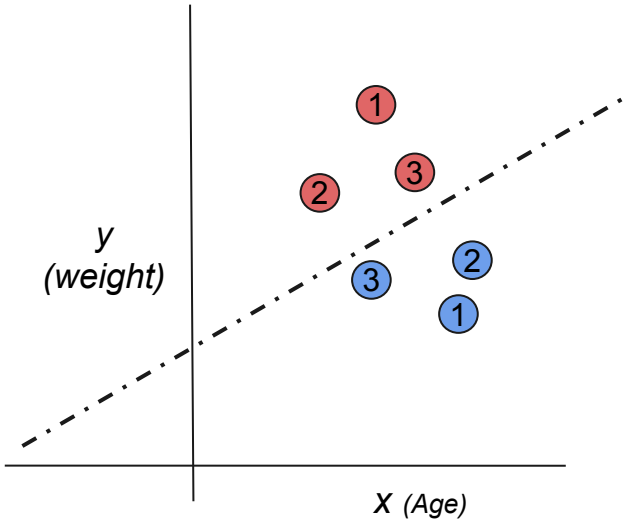



- Let's do feature encoding: ***Binary {0, 1}***
- We have to sacrifice color density (all from the same class have the same label, either 0/1

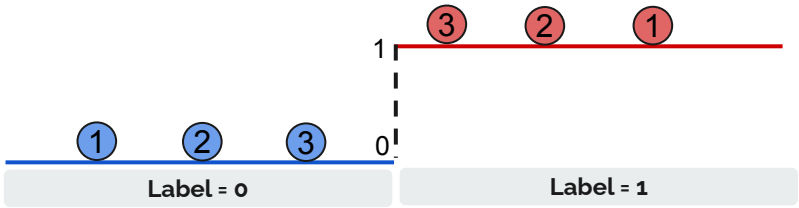


# Logistic Regression

## From Regression to Classification

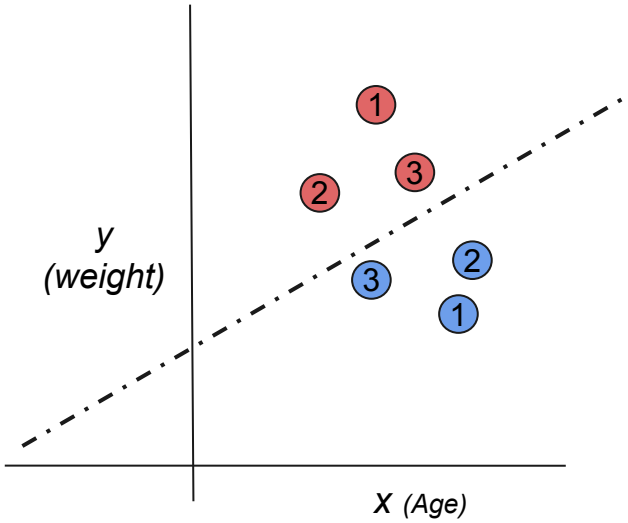



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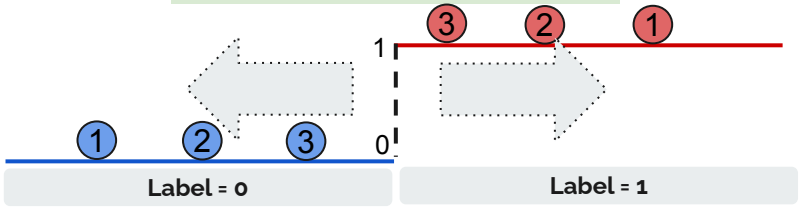


# Logistic Regression

## From Regression to Classification

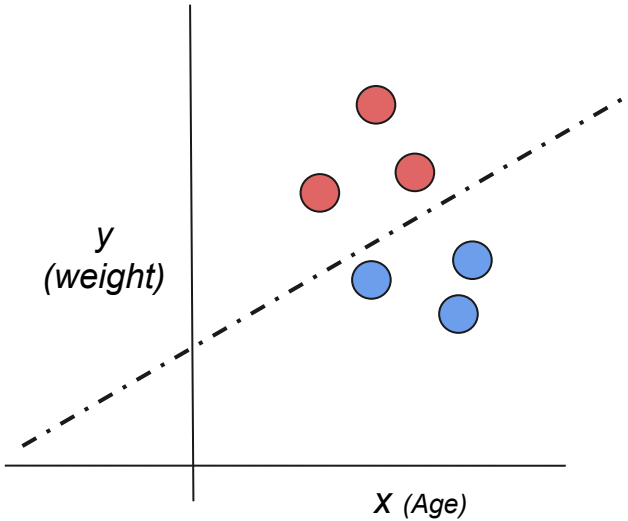



- Let's do feature encoding: ***Binary {0, 1}***
- We have to sacrifice color density (all from the same class have the same label, either 0/1
- We capture the indexing relationship with the distance from the 0-axis.



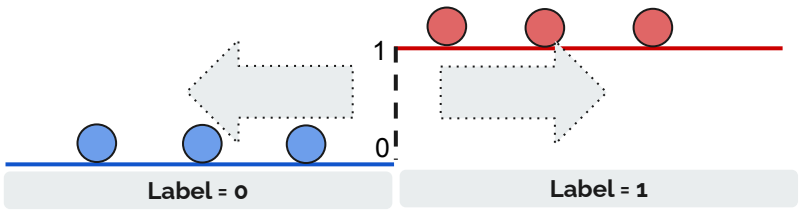
# Logistic Regression

## From Regression to Classification



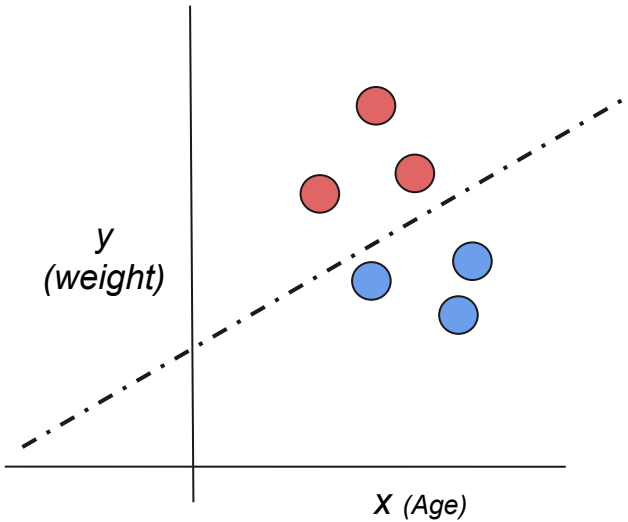

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so far are captured through the above function;*



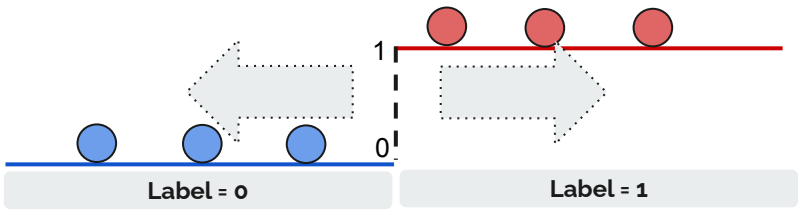
# Logistic Regression

## From Regression to Classification



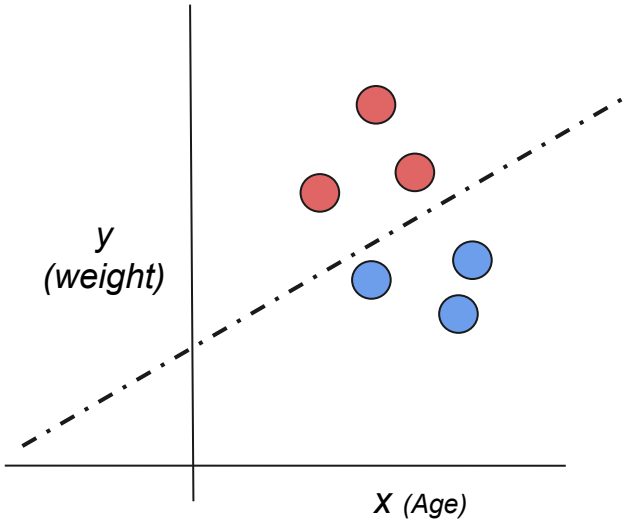

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

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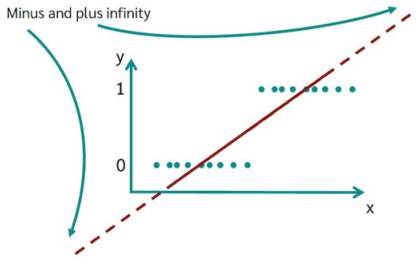
# Logistic Regression

## From Regression to Classification



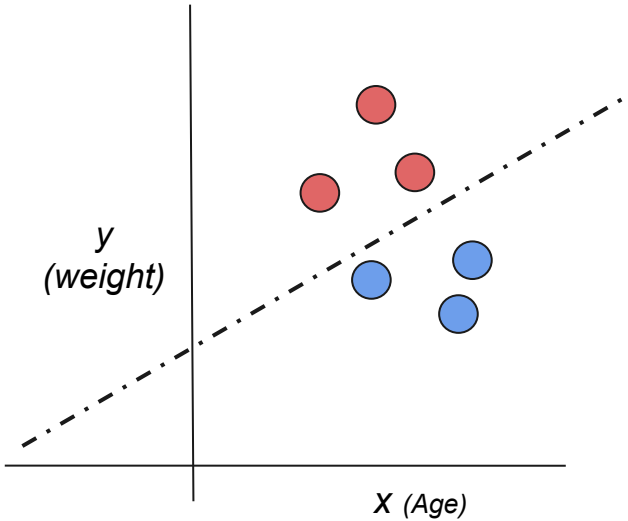

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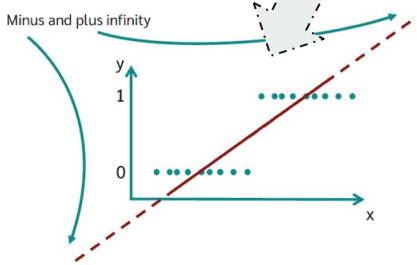
# Logistic Regression

## From Regression to Classification



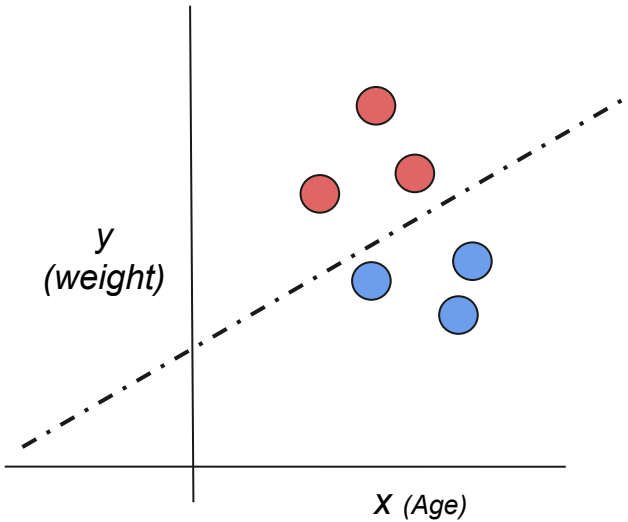

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so f[illegible]are captured through the above functi[illegible]how?*



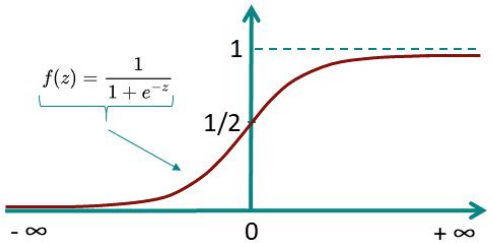
# Logistic Regression

## From Regression to Classification




$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so far are captured through the above function; how?*



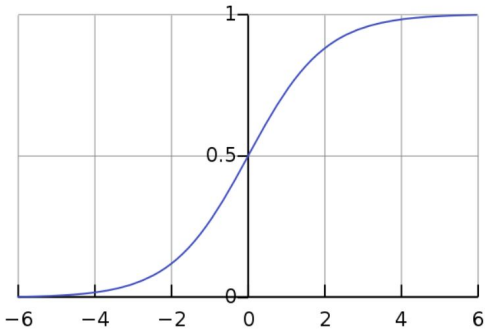
# Logistic Regression

- Probabilistic classifier

Sigmoid function characteristic

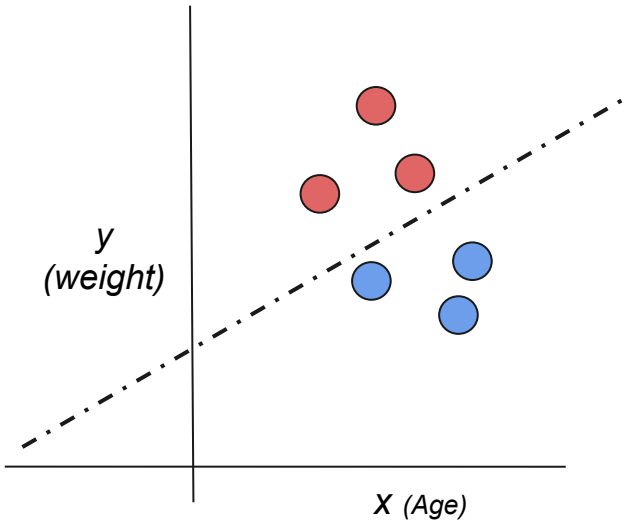
$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- Sigmoid function



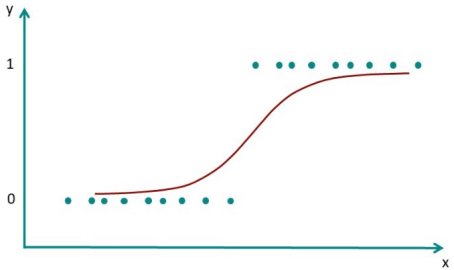
# Logistic Regression

## From Regression to Classification

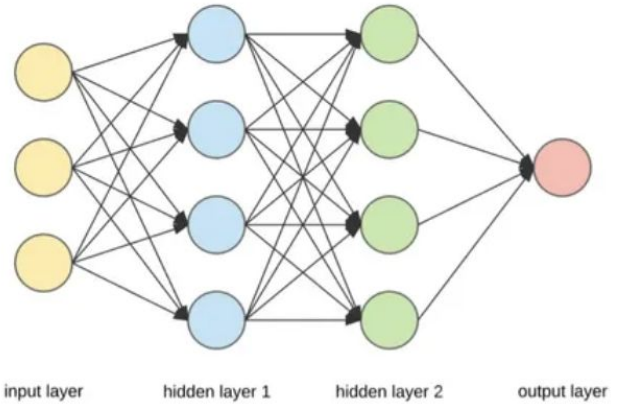



$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

- *All we have discussed so far are captured through the above function; how?*



# What's its connection to Neural Networks

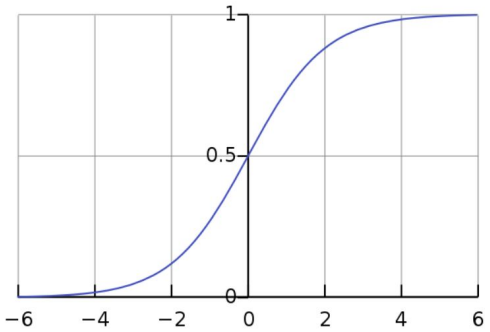


# Logistic Regression

- Probabilistic classifier

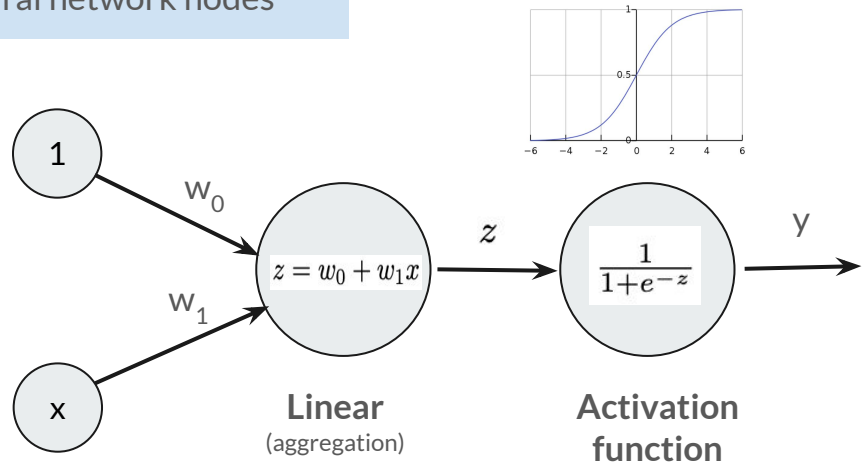
$$p(x) = \frac{1}{1+e^{-(w_0+w_1 x)}}$$

- Sigmoid function



# Logistic Regression to Neural Networks (NNs)

Concept of neural network nodes





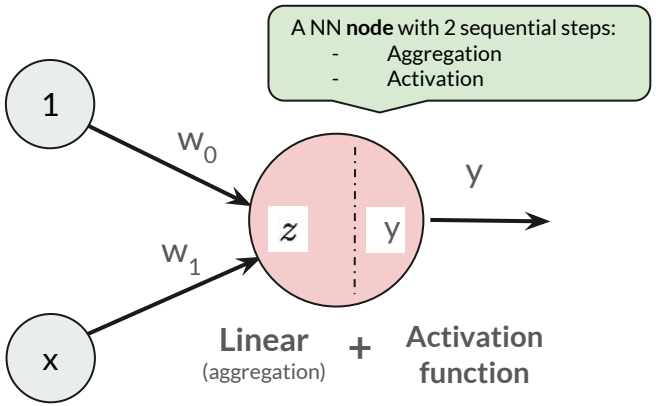


$y \in \{0, 1\}$

**Logistic Regression**

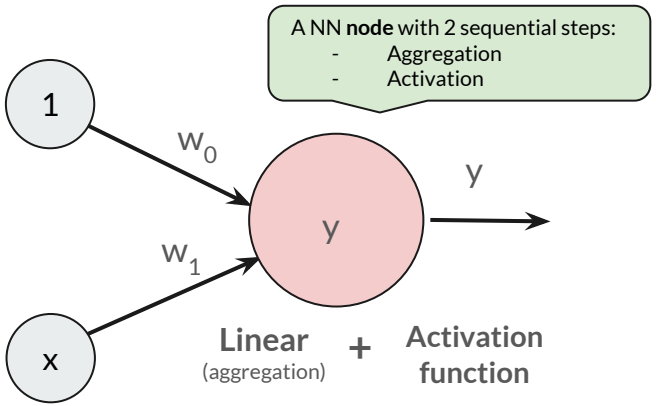
# Neural Networks (Node)

Concept of neural network nodes



# Neural Networks (Node)

Concept of neural network nodes



# QA